

Practical Management of Cardiovascular Disease in Chronic Kidney Disease

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PREVENTION AND CARDIOVASCULAR RISK REDUCTION

Control of traditional risk factors and other factors contributing to cardiovascular risk in chronic kidney disease (CKD) is critical to reducing morbidity and mortality burden in this high-risk population.

Lifestyle Factors and Smoking

Because physical inactivity is associated with albuminuria and cardiovascular mortality, CKD patients should be advised to be as physically active as possible and to avoid smoking.

Weight and Diet

Lifestyle changes, including balanced diets with regard to saturated fat and carbohydrates (in diabetic patients), probably reduce cardiovascular morbidity and should be encouraged. However, in all CKD stages, protein-energy wasting must be avoided (see [Chapter 90](#)). Especially in dialysis patients, increased body mass index is associated with better outcomes, possibly reflecting worse outcomes among underweight or normal weight patients who also have inflammation. Alternatively, obesity may act as a physiologic reserve against acute illness.

Hypertension

Blood pressure (BP) targets for CKD patients, in particular those with diabetes or proteinuria greater than 1 g/day, are discussed in [Chapter 82](#). Key goals of therapy include volume control (see [Chapter 82](#)) and prevention of sodium overload, particularly through dietary sodium restriction. Longer or more frequent hemodialysis (HD) sessions may be beneficial in controlling hypertension. Because of their cardioprotective effects, angiotensin-converting enzyme (ACE) inhibitors and angiotensin receptor blockers (ARBs) are the first-line drugs to treat hypertension in patients with CKD or end-stage kidney disease (ESKD). Diuretics, calcium channel blockers, and most other antihypertensive drugs, including centrally acting sympathetic inhibitors, are useful in combination with first-line agents, especially given that combination therapy is usually required to achieve recommended BP targets in CKD. Pure vasodilators (e.g., minoxidil) should be avoided because they may increase volume overload or occasionally cause pericardial effusion. Because sleep-disordered breathing occurs in about 50% of patients in CKD stages 4 to 5 and is associated with hypertension, sleep apnea should be considered in therapy-resistant hypertension.

Diabetes Mellitus

Optimal control of BP and glycemia (see [Chapter 32](#)) and treatment of dyslipidemia are crucial in managing diabetic CKD patients. Cardiovascular disease (CVD) should be treated aggressively in this high-risk group. Because the harm associated with severe hypoglycemia might counterbalance the potential benefit of intensive

glucose-lowering treatment, treating to a hemoglobin A_{1c} (HbA_{1c}) level less than 7.0% (53 mmol/mol) is not recommended in CKD stages 3 to 5.¹ Sodium-glucose cotransporter 2 (SGLT2) inhibitors and glucagon-like peptide-1 receptor agonists reduce cardiovascular events in patients with type 2 diabetes at high cardiovascular risk. Thus, various guidelines recommend these agents in CKD and non-CKD patients with CVD and/or multiple cardiovascular risk factors.²⁻⁴

Mineralocorticoid Receptor Antagonism

Mineralocorticoid receptor antagonists reduce the aldosterone-mediated proinflammatory effects that are involved in the fibrotic remodeling processes. The new selective nonsteroidal mineralocorticoid receptor antagonist finerenone also blocks the damaging effects of the overactivated aldosterone system. In contrast to the mineralocorticoid receptor antagonists spironolactone and eplerenone, finerenone is equally distributed in myocardial and kidney tissue. Finerenone binds to the same ligand domain but to different amino acids, leading to a different expression pattern of cardiac genes compared with spironolactone and eplerenone. Finerenone also reduced cardiac fibrosis and inflammation more than eplerenone in animal experiments at a comparable dose. In a large cardiovascular outcome trial (FIDELIO-DKD [Finerenone in Reducing CV Mortality and Morbidity in Diabetic Kidney Disease]) in type 2 diabetic patients with CKD, finerenone compared with placebo reduced the risk of kidney failure, cardiovascular death, nonfatal myocardial infarction (MI), nonfatal stroke, or hospitalization for congestive heart failure (CHF) with a low incidence of hyperkalemia.⁵ Based on the inclusion criteria of the trial, patients with diabetes and moderately increased albuminuria and impaired estimated glomerular filtration rate (eGFR) or severely increased albuminuria and eGFR should be treated with finerenone.

SGLT2 Inhibitor Treatment

SGLT2 inhibitors cause glucosuria and osmotic diuresis along with weight loss and natriuresis (see [Chapter 32](#)). In patients with diabetes, SGLT2 inhibitors significantly reduced cardiovascular morbidity and mortality. CREDENCE (Canagliflozin and renal outcomes in diabetic nephropathy) was the first study with an SGLT2 inhibitor in type 2 diabetic patients with CKD (eGFR ≥ 30 to < 90 mL/min/1.73 m²) with a combined primary kidney endpoint⁶: within 2.5 years, canagliflozin reduced the risk of kidney replacement therapy, doubling of serum creatinine and death due to kidney insufficiency by 33% versus placebo in addition to reductions of major cardiovascular events and hospitalization for CHF. The DAPA-CKD (Dapagliflozin And Prevention of Adverse outcomes in Chronic Kidney Disease) trial showed that in patients with nondiabetic CKD (eGFR 25–75 mL/min/1.73 m²), dapagliflozin reduced not only the primary endpoint of sustained decline in eGFR of 50% or more, ESKD, and kidney or cardiovascular

death but also the secondary endpoint of heart failure hospitalization and cardiovascular death.⁷

Dyslipidemia

Because CKD, like diabetes, is considered a cardiovascular risk equivalent, CKD patients should be treated to achieve the guideline goal for secondary prevention for low-density lipoprotein (LDL) cholesterol (<55 mg/dL) if eGFR is less than 30 mL/min/1.73 m² or less than 70 mg/dL in patients with CKD stage 3.⁴ Post hoc subgroup analyses in the Heart Protection Study, Cholesterol and Recurring Events (CARE) study and the Treating to New Targets (TNT) study support the use of statins in improving outcomes in CKD stages 1 to 2. The Study of Heart and Renal Protection (SHARP) showed that reducing LDL cholesterol with simvastatin 20 mg/day plus ezetimibe 10 mg/day safely reduced the incidence of major atherosclerotic events in patients with advanced CKD.⁸ However, the negative results from the 4D study and A Study to Evaluate the Use of Rosuvastatin in Subjects on Regular Hemodialysis: An Assessment of Survival and Cardiovascular Events (AURORA) do not support initiation of statins (with or without ezetimibe) in dialysis patients. Overall, the beneficial effect of lipid-lowering therapies is attenuated in subjects with low glomerular filtration rate (GFR) and very limited in ESKD patients on HD (Table 81.1).

Anemia

Partial correction of severe anemia with erythropoiesis-stimulating agents (ESAs) results in regression of left ventricular hypertrophy (LVH), and treatment of severe anemia is also associated with fewer ischemic symptoms in coronary heart disease patients. However, randomized controlled trials in HD and CKD patients have shown no benefit of ESA treatment on mortality (see Chapter 86). Thus, anemia correction in patients with advanced CKD to a hemoglobin level of 10 to 12 g/dL is recommended (www.kdigo.org).

Inflammation

A careful search for infectious processes, such as periodontal disease, is recommended in dialysis patients with inflammation. Restriction of central venous catheter use is also important; short daily dialysis with better fluid status was associated with decreasing C-reactive protein (CRP) levels compared with conventional HD. Volume status should be carefully monitored to avoid inflammation. Altered intestinal microbial flora as a potential risk factor for systemic uremic inflammation merits further study. Medium cutoff dialysis membranes reduce uremic inflammation,⁹ but larger trials with longer treatment periods are encouraged. Interestingly, the CANTOS trial (Canakinumab Antiinflammatory Thrombosis Outcome Study) demonstrated that inhibiting proinflammatory effector molecule interleukin-1 β (IL-1 β) with canakinumab reduced the risk of cardiovascular events, and that the benefit appeared larger in patients with eGFR less than 60 mL/min/1.73 m² than in those with eGFR greater than or equal to 60 mL/min/1.73 m².¹⁰

Chronic Kidney Disease–Mineral Bone Disorder

Recent meta-analyses and the updated Kidney Disease: Improving Global Outcomes (KDIGO) chronic kidney disease–mineral bone disorder (CKD-MBD) guidelines (www.kdigo.org) suggest that avoiding calcium-containing phosphate binders may reduce cardiovascular events, calcification progress, and mortality; however, there is a low level of evidence and multiple studies have not confirmed this (see Chapter 88). We suggest that these agents should be avoided in CKD patients with significant life expectancy, particularly in patients on the transplant waiting list, independent of the presence or absence of calcifications. The potential detrimental effects of sevelamer (and other commonly used drugs) on gut microflora require further studies.¹¹

Results from the Dialysis Outcomes and Practice Patterns Study (DOPPS) call into question the survival advantage for HD patients taking vitamin D. In the Paricalcitol Capsules Benefits Renal Failure Induced Cardiac Morbidity in Subjects with CKD Stage 3/4 (PRIMO) study, 48 weeks of paricalcitol therapy did not alter cardiac mass index or improve measures of diastolic dysfunction.¹² Thus, the use of native vitamin D or vitamin D receptor agonists in CKD patients should primarily be governed by CKD-MBD parameters (see Chapter 88).

The Evaluation of Cinacalcet HCl Therapy to Lower Cardiovascular Events (EVOLVE) study¹³ showed that cinacalcet only reduced the risk for death or major cardiovascular events in dialysis patients with moderate to severe secondary hyperparathyroidism after adjustment for age—a nominally significant finding. In a second trial in similar dialysis patients, cinacalcet plus low-dose vitamin D sterols tended to attenuate vascular and cardiac valve calcification compared with an exclusively vitamin D sterol-based approach.¹⁴

FGF-23, a bone hormone, is cardiotoxic and induces LVH but does not induce vascular calcification. In a post hoc analysis of EVOLVE data, treatment of secondary hyperparathyroidism with cinacalcet was associated with reduced FGF-23 levels, and patients with a 30% or 50% drop in FGF-23 levels experienced significantly fewer cardiovascular events.¹⁵ In patients with advanced secondary hyperparathyroidism, treatment with a calcimimetic alone may therefore be preferred to combined therapy with a vitamin D agonist.

CORONARY ARTERY DISEASE IN PATIENTS WITH CHRONIC KIDNEY DISEASE

With respect to the diagnosis of coronary artery disease (CAD), noninvasive stress testing shows reduced accuracy in patients with CKD and in particular in those on HD. Both medical and interventional therapy of acute or chronic coronary syndromes are not different between patients with or without CKD, and patients with CKD should receive a similarly stringent treatment with drugs/procedures that have been shown to provide a prognostic benefit.

Chronic Coronary Syndrome

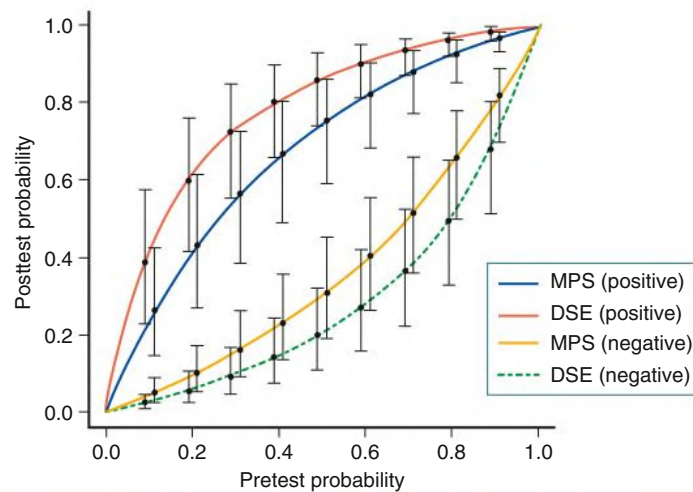
Diagnosis

CAD in patients with CKD differs from nonrenal controls by exhibiting more and larger plaques and more signs of inflammation.¹⁶ Dialysis patients also exhibited more calcified plaques in coronary arteries, whereas plaques of nonrenal patients were mostly fibro-atheromatous.¹⁷ In addition, a lower eGFR was associated with increased numbers of newly formed intramural blood vessels and intraplaque hemorrhages,^{17,18} suggesting a more vulnerable plaque phenotype.

In CKD patients with suspected obstructive CAD, special attention should be paid to the fact that symptomatic angina is less common, whereas silent ischemia is more common.¹⁹ The diagnostic approach to CAD in CKD patients is similar compared with non-CKD patients,⁴ except that accuracy of noninvasive testing is lower, particularly in dialysis patients.²⁰ The National Kidney Foundation's Kidney Disease Outcomes Quality Initiative (KDOQI) clinical practice guidelines for CVD in dialysis patients recommend an electrocardiogram at dialysis initiation and then annually.²¹

In addition, KDOQI guidelines recommend echocardiography in all dialysis patients after they achieve “dry weight” targets, preferably 1 to 3 months after dialysis initiation on an interdialytic day for HD patients and at 3-year intervals thereafter.²² The rationale is that diminished left ventricular (LV) systolic function, an important independent risk factor for CVD and mortality,²³ is not accurately diagnosed

Accuracy of DSE vs MPS for Diagnosing CAD in Kidney Transplant Recipients



Posttest Probability			
Test	Pretest probability (%) of coronary artery disease	Posttest probability (%) after positive result	Posttest probability (%) after negative result
Dobutamine stress echocardiography (DSE)	Low risk (10-29)	42-72	3-10
	Intermediate risk (30-59)	73-90	10-27
	High risk (60-90)	91-98	28-70
Myocardial perfusion scintigraphy (MPS)	Low risk (10-29)	24-54	5-15
	Intermediate risk (30-59)	55-81	16-38
	High risk (60-90)	81-96	39-79

Fig. 85.1 Pre test and post test probabilities of coronary artery disease (CAD) associated with positive and negative results of dobutamine stress echocardiography (DSE) versus myocardial perfusion scintigraphy (MPS) in renal kidney transplant candidates. (From Shroff GR, Herzog CA. Coronary revascularization in patients with CKD stage 5D: pragmatic considerations. *J Am Soc Nephrol*. 2016;27:3521–3529.)

by history, physical examination, or chest radiography. Detection of unsuspected cardiomyopathy is also important, given that carvedilol therapy in such patients improved LV systolic function, decreased hospitalization, and reduced mortality. As in the general population, CKD patients with left ventricular ejection fraction (LVEF) below 40% should be evaluated for coronary heart disease (exceptions are pediatric or young adult patients with nondiabetic CKD and other patients known to be at low risk for CAD).

Overall, noninvasive stress testing shows reduced accuracy (vs. invasive coronary angiography) in patients with advanced CKD²⁴ compared with patients with normal kidney function. The accuracy of both pharmacologic stress echocardiography and single-photon emission computed tomographic nuclear imaging is lowest in ESKD patients even under the most optimal circumstances (i.e., in clinics with a high level of expertise). Reviews cite overall sensitivity percentages in the mid-70s and specificity in the mid-80s for stress echocardiography and sensitivity and specificity in the mid-70s for SPECT myocardial perfusion imaging. Based on Medicare data, 87% of cardiac stress testing in US patients with CKD employs nuclear scintigraphy.²⁵

ESKD patients are poorly suited for conventional exercise stress electrocardiography because of limited exercise tolerance and frequent resting electrocardiographic abnormalities. As noted earlier, even under optimal circumstances, the most widely used stress imaging

techniques (compared with invasive coronary angiography) are at best imperfect, and in real-world clinical practice they may even be worse. For this reason, local institutional expertise with imaging of ESKD patients will dictate clinical practice irrespective of the published “theoretical” accuracy. Moreover, prediction of the likelihood of future events may differ considerably from predictions based on coronary anatomy. Noninvasive stress testing is probably better at predicting mortality than coronary anatomy—something that continues to vex clinicians, as typically in an individual patient we don’t want to miss “significant” obstructive CAD (i.e., angiographic definition), yet from the broader perspective of population epidemiology, the focus is on events like MI or death (not angiographic stenoses). A meta-analysis concluded that presence of inducible myocardial ischemia by any stress-imaging test is independently predictive of increased acute MI risk and cardiac death, whereas a fixed or resting defect or abnormality is predictive of cardiac death but not acute MI.²⁶

In kidney transplant candidates the sensitivities and specificities of stress nuclear and echocardiographic imaging for detection of coronary heart disease have been reported to range from 44% to 90%—not terribly helpful to the nephrologist or even cardiologist dealing with an individual patient. Consistent with others,²⁵ a meta-analysis (Fig. 85.1) concluded that dobutamine stress echocardiography is probably more accurate than myocardial stress nuclear scintigraphy for noninvasive

Management of Coronary Artery Disease in Renal Transplant Candidates

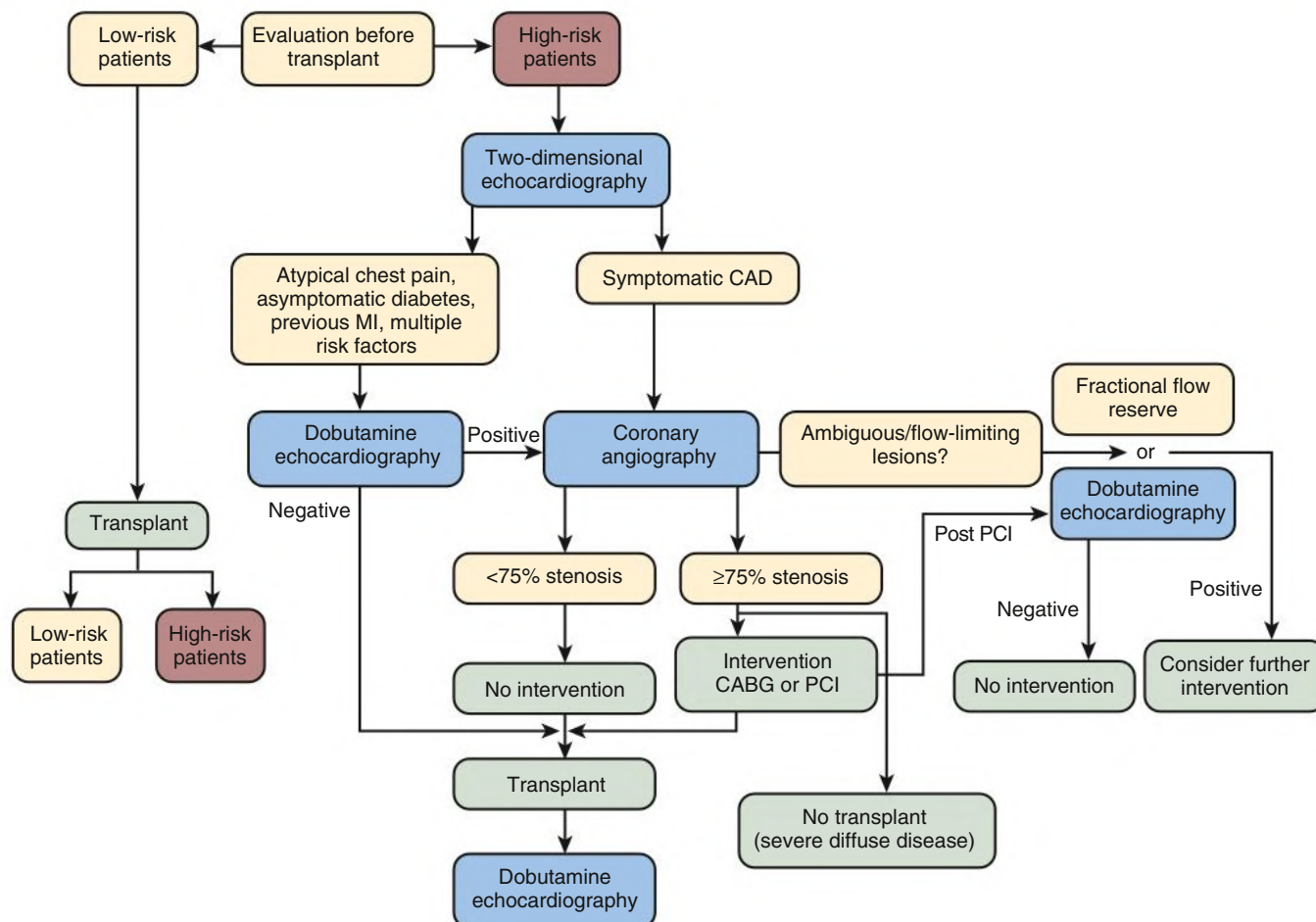


Fig. 85.2 Algorithm for management of coronary heart disease in kidney transplant candidates. CABG, Coronary artery bypass graft surgery; CAD, coronary artery disease; MI, myocardial infarction; PCI, percutaneous coronary intervention. (Modified from Herzog CA. Acute MI in dialysis patients: how can we improve the outlook? *J Crit Illn*. 1999;14[11]:613–621.)

detection of coronary heart disease in kidney transplant candidates.²⁷ However, a recent analysis concluded “there are no compelling reasons to routinely select one particular stress imaging modality in preference to others, although there may specific circumstances that favor a particular test.”²⁵

The major problem with searching for CAD in asymptomatic kidney transplant candidates (or dialysis patients) is that the evidence for prophylactic revascularization is weak.²¹ The best observational data supporting preemptive coronary revascularization showed a 3-year cardiac event–free survival of 90% for transplant wait-listed revascularized patients.²⁸ Optimal medical therapy (which should constitute the treatment strategy for all patients) may potentially attenuate the putative benefit of prophylactic coronary revascularization. One algorithm for screening and management of CAD in kidney transplant candidates is presented in Fig. 85.2. However, challenging the rationale for this or any other algorithm employing prophylactic coronary revascularization, in the ISCHEMIA-CKD trial (see later) an invasive strategy of optimal medical therapy and coronary angiography (with revascularization) did not improve outcomes compared with optimal medical therapy alone.²⁹

Biomarkers

Plasma brain natriuretic peptides (BNP and NT-proBNP), cardiac troponins (cTnT, cTnI), and high-sensitivity CRP (hsCRP) are prognostic risk markers in the evaluation of heart disease in ESKD.³⁰ BNP reflects cardiac filling pressures (not limited to the left heart), troponins reflect myocardial cell death (but not necessarily ischemia), and hsCRP reflects inflammation. Elevation of cTnT occurs even in pediatric CKD patients and is associated with cardiac dysfunction. Elevated levels of serum troponin in ESKD patients should not be uncritically attributed to myocardial ischemia caused by obstructive CAD as they also reflect the presence and severity of HD-induced myocardial stunning.³¹ The cardiac biomarker–based diagnosis of acute coronary syndrome requires a time-appropriate rise and fall of the biomarker. Fig. 85.3 graphically displays the relationship of cTnT and cTnI levels in asymptomatic dialysis patients and long-term survival. On the basis of these data, the US Food and Drug Administration approved the measurement of cTnT in dialysis patients for risk stratification (mortality prediction).

High-sensitivity cardiac troponin (hs-cTn) assays are highly precise and serve two complementary roles: the diagnosis of acute MI

Survival by Baseline Troponin Values

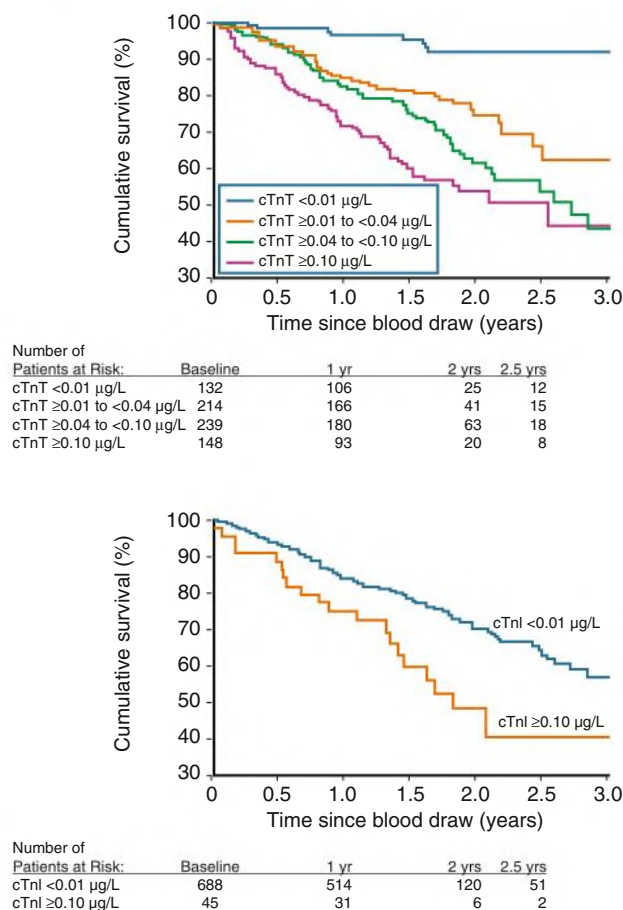


Fig. 85.3 Kaplan-Meier survival curves by baseline troponin values. *cTnI*, Cardiac troponin I; *cTnT*, cardiac troponin T. (From Apple FS, Murakami MM, Pearce LA, Herzog CA. Predictive value of cardiac troponin I and T for subsequent death in end-stage renal disease. *Circulation*. 2002;106:2941–2945.)

and risk stratification based on detection of hs-cTn in asymptomatic patients (e.g., perhaps a marker for apoptosis). Similar to elevated troponin serum levels discussed earlier, hs-cTn-based risk stratification should also employ changes in values over time, rather than simple thresholds.³²

Coronary Angiography

Coronary angiography should be considered in stable ESKD patients with evidence for inducible myocardial ischemia, unstable patients with acute coronary syndrome (performed urgently for ST-segment elevation MI [STEMI]), and patients with LVEF less than 40%. In CKD and dialysis patients with residual kidney function, fear of contrast nephropathy may restrain use of coronary angiography (see Chapter 74 for preventive measures). However, one retrospective study of 76 nondialysis patients with mean eGFR of 12.5 mL/min/1.73 m² found no significant postangiographic deterioration in kidney function. Although acute kidney injury (AKI) after coronary angiography increases the risk of mortality, ESKD, and hospitalization,³³ fear of AKI should not deter clinically mandated coronary angiography. Echocardiography should be performed before any nonemergent

coronary angiography in CKD patients to diagnose clinically unsuspected valvular disease or cardiomyopathy, to gauge preprocedure volume status, and to assess LV function (to avoid excessive exposure to radiocontrast media through unwarranted ventriculography).

Noninvasive coronary CT angiography may be problematic in dialysis patients because of medial calcification interfering with angiographic interpretation.³⁴ Coronary CT angiography is noninvasive and has similar accuracy to invasive coronary angiography but is associated with a lower risk of significant AKI in kidney transplant candidates. Noninvasive gadolinium-based magnetic resonance angiographic imaging in patients with severe CKD remains problematic because of residual concerns about nephrogenic fibrosing dermopathy (see Chapter 91).

Drug Therapy in Chronic Coronary Syndrome Patients With Chronic Kidney Disease

According to current guidelines, drug therapy in chronic coronary syndrome (CCS) patients with CKD should not differ from therapy in non-CKD patients, but renally excreted drugs used in CCS should be dose adjusted for kidney function.

Revascularization

CKD is a strong risk factor for death after coronary revascularization with percutaneous coronary intervention (PCI) or coronary artery bypass grafting (CABG).^{35,36} Coronary revascularization complicated by AKI is associated with excess mortality; operative mortality for non-CKD patients who require acute dialysis after cardiac surgery is 44%.

Observational studies suggest lower mortality in CKD (including dialysis) patients undergoing coronary revascularization compared with no revascularization. A post hoc analysis of CKD patients in the Fast Revascularization During Instability in Coronary Artery Disease (FRISC II) trial indicated a superior outcome with an early invasive strategy in ACS compared with conservative management. However, the optimal coronary revascularization method in CKD remains controversial. A post hoc analysis of CKD patients enrolled in the Arterial Revascularization Therapies Study (ARTS) found similar outcomes for CABG or multivessel PCI with non-drug-eluting stents for death, MI, or stroke. In elderly non-dialysis-dependent CKD patients, the incidence of ESKD was lower after PCI (5.4% at 3 years vs. 6.8% for CABG), but long-term risk for death (3-year mortality with PCI 33% vs. 28% with CABG) or the combined event of death or ESKD was lower after CABG. The relative survival advantage of CABG versus PCI occurs only more than 6 months after revascularization.³⁷ Dialysis patient survival after CABG is better than after PCI with non-drug-eluting stents³⁸ and drug-eluting stents,³⁹ but 2-year mortality remains high at 44% versus 52% for PCI.³⁸ In the absence of data from dedicated cardiovascular outcome trials in patients with ESKD or on HD, recommendations are based on large observational studies. Current guidelines recommend that the revascularization strategy (PCI vs. CABG) should be chosen based on the general condition, preferences, and life expectancy of the patient, the least invasive approach being more appropriate in the most fragile and compromised patients.⁴⁰

The ISCHEMIA-CKD trial compared an invasive or conservative care approach in CKD patients (eGFR <30 mL/min/1.73 m² or dialysis) with stable CAD. Patients with moderate or severe ischemia on stress testing were randomly assigned to an initial invasive strategy consisting of coronary angiography and revascularization (if appropriate) added to medical therapy or an initial conservative strategy consisting of medical therapy alone and angiography reserved for those in whom medical therapy failed. After a median follow-up of 2.2 years, there was no difference for the primary composite endpoint of death or nonfatal MI between groups. Compared with the conservative strategy, the invasive strategy was associated with a significantly

Estimated Mortality of Dialysis Patients After Acute Myocardial Infarction

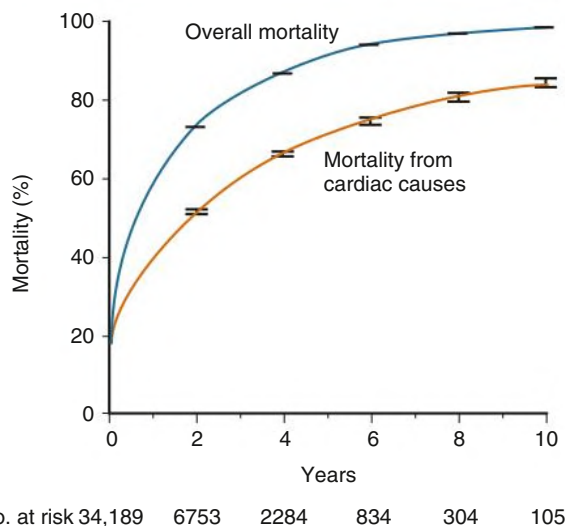


Fig. 85.4 Estimated mortality of dialysis patients after acute myocardial infarction. (From Herzog CA, Ma JZ, Collins AJ. Poor long-term survival after acute myocardial infarction among patients on long-term dialysis. *N Engl J Med* 1998;339:799–805.)

higher incidence of stroke and a higher incidence of death or initiation of dialysis.²⁹ In addition, both groups did not differ with respect to angina-related health status during follow-up.⁴¹

Acute Coronary Syndrome

Acute MI in dialysis patients is associated with poor long-term survival. Unadjusted 2-year mortality has not improved over time (71% in 1977–1984 and 72% in 2008⁴²) despite dramatic improvements in acute MI outcomes in the general population (Fig. 85.4). Significant improvement has occurred only in dialysis patients with STEMI, specifically related to in-hospital mortality. Rates of non-STEMI (NSTEMI) mortality or postdischarge mortality have not improved.⁴² In-hospital mortality increases with decreasing GFR, which has been attributed to underrecognition resulting from atypical presentations, underuse of appropriate diagnostic investigations, and undertreatment (therapeutic nihilism).⁴³ A US registry of dialysis patients hospitalized for acute MI found the following⁴³:

- With respect to acute coronary syndrome, 45% of dialysis patients were diagnosed incorrectly versus 21% of nondialysis patients.
- ST-segment elevation was found in 19% of dialysis patients versus 36% of nondialysis patients.
- After other clinical exclusions, 10% of dialysis patients and 25% of nondialysis patients were eligible for acute coronary reperfusion.
- Of those who were eligible, 47% of dialysis patients and 75% of nondialysis patients actually received reperfusion.

Similar findings occur across the CKD spectrum; the likelihood of increased mortality and lower prevalence of both STEMI and chest pain are correlated with severity of non-dialysis-dependent CKD in patients with acute MI.^{44,45} Compared with those with normal eGFR, patients with eGFR less than 45 mL/min/1.73 m² are three times as likely to present with acute MI as the initial manifestation of coronary heart disease, rather than stable angina.

Given the poor prognosis of patients with acute MI and the fact that CKD patients are less likely to receive appropriate therapy, current guidelines recommend that acute coronary syndrome patients with CKD should be treated as aggressively as in patients without CKD.

HEART FAILURE IN PATIENTS WITH CHRONIC KIDNEY DISEASE

Patients with CKD show an increased risk to develop heart failure, and the presence of this comorbidity tremendously impairs the prognosis of CKD patients. Some therapeutic approaches in heart failure in patients have not been tested in patients with advanced CKD. However, SGLT2 inhibitors have shown benefit in both patients with heart failure and CKD and should be used in CKD patients not on HD (see Chapter 82).

Heart failure in patients with CKD, in particular those on HD, is possibly related to maladaptive ventricular hypertrophy and fibrosis.⁴⁶ LVH occurs early in progressive CKD, probably because of high hypertension prevalence, including frequent nocturnal hypertension. Pressure overload, caused by hypertension and arterial stiffness, results in concentric hypertrophy. Volume overload manifests as eccentric hypertrophy. Left ventricular dilation strongly predicts poor outcome. It may be an end result of severe LVH, diffuse ischemic damage, or recurrent volume overload; a high-output arteriovenous fistula may contribute. Diastolic dysfunction is strongly associated with LVH and risk for intradialytic hypotension.

Hemodialysis can produce repetitive myocardial injury, leading to global and segmental reduction in left ventricular systolic function; HD-induced myocardial stunning is associated with increased 1-year mortality. Even pediatric HD patients can experience HD-induced myocardial stunning, indicating that large-vessel obstructive CAD is not a prerequisite for this pathologic finding. Biofeedback dialysis or reduced dialysate temperature can help reduce intradialytic hypotension and severity of HD-induced myocardial stunning.

With respect to HF diagnosis, NT-proBNP and BNP are equivalent predictors of decompensated heart failure in CKD, but NT-proBNP is a better predictor of survival.

Congestive Heart Failure Treatment in Patients With Chronic Kidney Disease

No treatments convincingly reduce morbidity and mortality in CKD patients with CHF and preserved ejection fraction (HFpEF; LVEF ≥50%) or moderately impaired left ventricular function (HFmrEF; LVEF 40%–49%).

Therapy of Heart Failure and Reduced Ejection Fraction in Chronic Kidney Disease

Treatment of CHF in the general population is based on data from large cardiovascular outcome trials that have led to the development of certain treatment algorithms.⁴⁷ However, patients with severe CKD were excluded from the landmark trials, and therefore recommendations for this group are largely based on data from the general population (Fig. 85.5).⁴⁸

ACE Inhibitors/Angiotensin 2 Receptor Blockers

ACE-inhibitors reduce morbidity and mortality in patients with heart failure, and reduced ejection fraction (HFrEF) and solid data exist for the use of ACE-inhibitors in patients with CKD stage I–III plus HFrEF, but no data exist for those with CKD stages 4 to 5. In patients with ACE-inhibitor intolerance it is recommended to treat with an angiotensin receptor blocker (ARB).

β-Blockers

First-line therapy in patients with HFrEF also includes treatment with β-blockers. A meta-analysis of six studies analyzing the effect of β-blocker therapy in CHF and CKD stages 3 to 5 show that patients with advanced CKD benefited from β-blocker therapy.⁴⁹

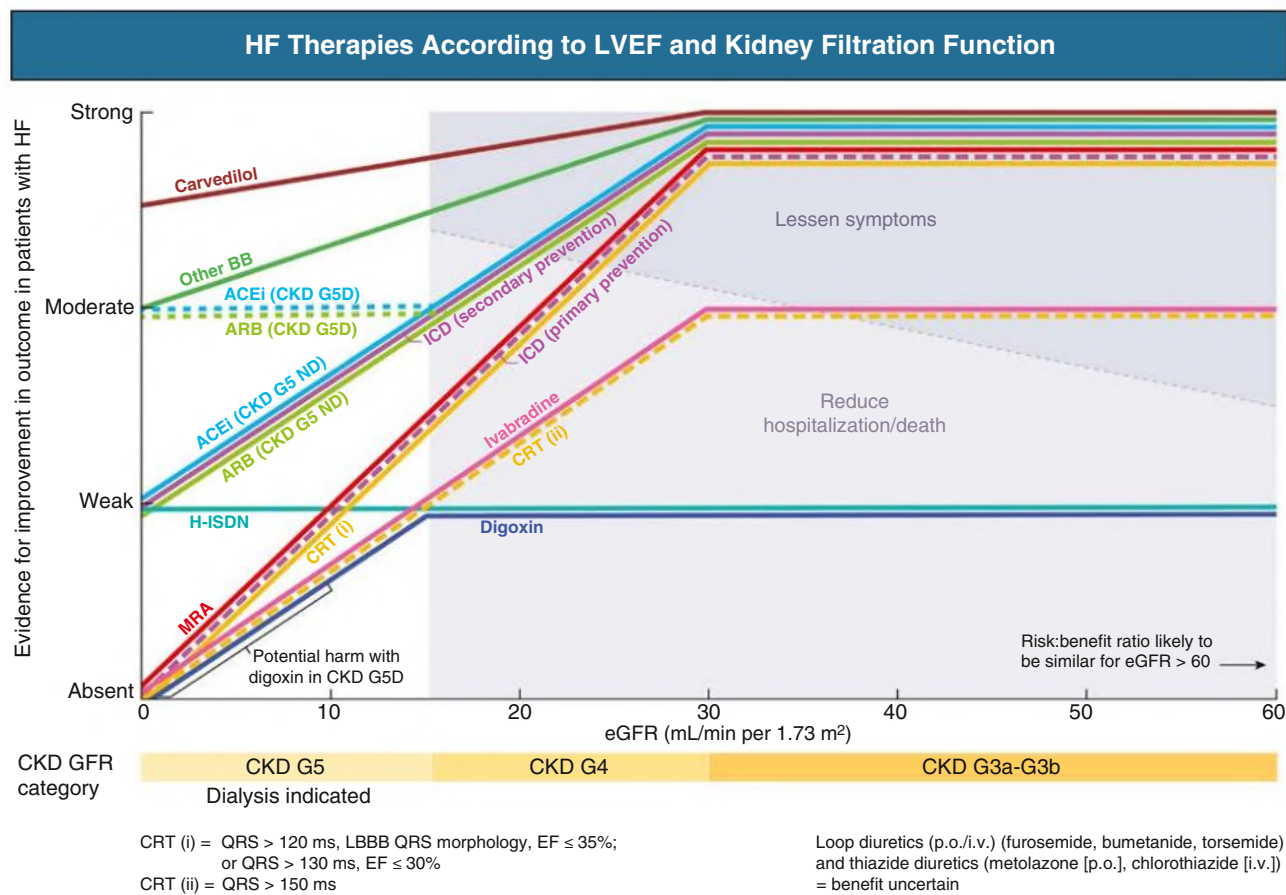


Fig. 85.5 Positioning of heart failure (HF) therapies according to left ventricular ejection fraction (LVEF) and glomerular filtration rate. ACEi, Angiotensin-converting enzyme inhibitor; ARB, angiotensin II receptor blocker; BB, β -blocker; CKD G5D, chronic kidney disease glomerular filtration rate category 5 patient on dialysis; CKD G5 ND, chronic kidney disease glomerular filtration rate category 5 patient not on dialysis; CRT, cardiac resynchronization therapy; EF, ejection fraction; eGFR, estimated glomerular filtration rate; GFR, glomerular filtration rate; H-ISDN, hydralazine and isosorbide dinitrate; ICD, implantable cardioverter-defibrillator; LBBB, left bundle branch block; MRA, mineralocorticoid receptor antagonist. (From House AA, Wanner C, Sarnak MJ, et al. Heart failure in chronic kidney disease: conclusions from a Kidney Disease: Improving Global Outcomes (KDIGO) Controversies Conference. *Kidney Int.* 2019;95:1304–1317.)

Mineralocorticoid Receptor Antagonists

If patients with HFrEF and a medication of ACE inhibitors and β -blockers are still symptomatic, and if LV ejection fraction is 35% or less, they should be treated with a mineralocorticoid receptor antagonist with attention in particular to the development of hyperkalemia. These agents are effective in patients with CHF and CKD stages 1 to 3⁵⁰ but are contraindicated in CKD stages 4 to 5.

Diuretics

Treatment with diuretics is indicated in patients with venous congestion and New York Heart Association (NYHA) II, as well as in patients with CHF stage NYHA III and IV. Patients can be treated with loop diuretics or thiazides, but patients with CKD stages 4 to 5 should receive either loop diuretics alone or thiazides in combination with loop diuretics. In particular, the latter should be monitored for hypokalemia. After cardiac decompensation, at least a low-dose diuretic therapy should be continued.

Angiotensin Receptor Neprilysin Inhibitor

If patients treated with a combination of ACE inhibitors, β -blockers, and mineralocorticoid receptor antagonists are still symptomatic, and if ACE inhibitors have been well tolerated, it is recommended to change from ACE inhibitors or ARBs to angiotensin receptor neprilysin inhibitor (ARNI)

treatment provided that kidney function permits. For LCZ696 (sacubitril + valsartan), a reduction of mortality, cardiovascular mortality, and CHF hospitalization has been shown in comparison to enalapril, and this benefit has also been found in patients with eGFR of 30 to 60 mL/min/1.73 m².⁵¹ Therefore, ARNIs are effective in patients with heart failure and CKD stages 1 to 3; in patients with CKD stages 4 to 5, no data are available.

SGLT2 Inhibitors

Two large cardiovascular outcome trials conducted in patients with HFrEF with or without diabetes have shown that dapagliflozin or empagliflozin significantly reduce the combined endpoint of cardiovascular death and CHF hospitalization compared with placebo. In addition, these agents reduce kidney endpoints and worsening of nephropathy. Given that these studies included patients with eGFR as low as 30 mL/min/1.73 m² (DAPA-HF)⁵² or 20 mL/min/1.73 m² (EMPEROR-Reduced),⁵³ these agents seem to be effective in patients with CKD stages 3 to 4. Based on these data, one of the SGLT2-inhibitors, dapagliflozin, has recently been approved for use in patients with HFrEF and eGFR as low as 30 mL/min/1.73 m² independent of the presence of diabetes.

Extracellular Volume Overload

Extracellular volume overload resulting from loss of sodium excretory capacity is the major cause of hypertension in dialysis patients.

Whether prevention of recurrent hypervolemia reduces cardiovascular morbidity and mortality remains unproven. If adjustments are made for comorbidity and advanced age, a strong, incremental risk for all-cause and cardiovascular mortality is associated with interdialytic weight gains. Recurrent hypervolemia may result in LVH and dilation, peripheral or pulmonary edema, jugular venous distension, or a third heart sound, or it may be largely asymptomatic. Tolerance of large ultrafiltration volumes intradialytic or increase in blood pressure may indicate that the dry weight target (see Chapter 99) has not been reached. Reaching the optimal dry weight does not necessarily lead to normotension; a lag phase of some weeks can precede improvement.

ARRHYTHMIAS IN PATIENTS WITH CHRONIC KIDNEY DISEASE

Atrial Fibrillation

Atrial fibrillation (AF) is the most common arrhythmia in advanced CKD, and its incidence is increasing in older patients initiating dialysis in the United States.⁵⁴ Incident AF is an independent risk factor for development of ESKD in CKD.⁵⁵ The historically reported prevalence among dialysis patients is 15% to 20%, but small series employing implantable loop recorders suggest that the true prevalence may be as high as 27% to 40%, if episodes of paroxysmal AF are included.⁵⁶

No trials have examined the clinical benefits and risks of anticoagulation in CKD populations specifically. In observational studies, adjusted-dose warfarin was associated with a 76% reduction in the relative risk for ischemic stroke or systemic embolism among AF patients with CKD stage 3.⁵⁷ Irrespective of CKD severity, in nondialysis acute MI patients with AF, warfarin treatment was associated with a lower 1-year risk for the composite outcome of death, MI, and ischemic stroke without a higher risk for bleeding. In contrast, the usefulness of warfarin

for primary prevention of stroke in dialysis patients with AF is controversial; some observational studies suggest harm, and a consensus statement from KDIGO did not advise routine warfarin therapy for primary prevention of stroke in AF patients with CKD stage 5D (however, warfarin for secondary prevention of stroke was suggested based on expert opinion).^{58,59} Warfarin use in dialysis patients is also challenging given the difficulty of maintaining the optimal anticoagulation range and the potential for accelerated vascular calcification with vitamin K antagonists. Post hoc analyses from randomized controlled trials suggest beneficial effects of direct oral anticoagulants compared with warfarin in moderate CKD (Fig. 85.6); however, in dialysis patients, treatment with apixaban was not associated with a lower incidence of new stroke, transient ischemic attack, or systemic thromboembolism but was associated with a higher incidence of fatal or intracranial bleeding.⁶⁰

No data support the sole use of aspirin for prevention of stroke in CKD patients with AF.

Sudden Cardiac Death

In the US Renal Data System database, 85% of all cardiac deaths and 34% of all-cause mortality in HD patients were attributable to arrhythmias; although overall cardiovascular and all-cause mortality have improved over time, deaths attributable to arrhythmic mechanisms have not, leading to a larger proportion of mortality attributable to arrhythmia (Fig. 85.7).⁶¹ The reported rate of cardiac arrest in HD centers is 3.8 to 7.1 events per 100,000 dialysis sessions. Strong predictors of sudden cardiac death are a history of coronary heart disease, peripheral arterial disease, diabetes, elevated inflammatory biomarkers, and reduced LVEF. Even a modest reduction in LVEF to 40% to 50% is prognostically important in both HD and peritoneal dialysis (PD) patients. Factors probably contributing to the special vulnerability of ESKD patients to sudden cardiac arrest include LVH, rapid electrolyte

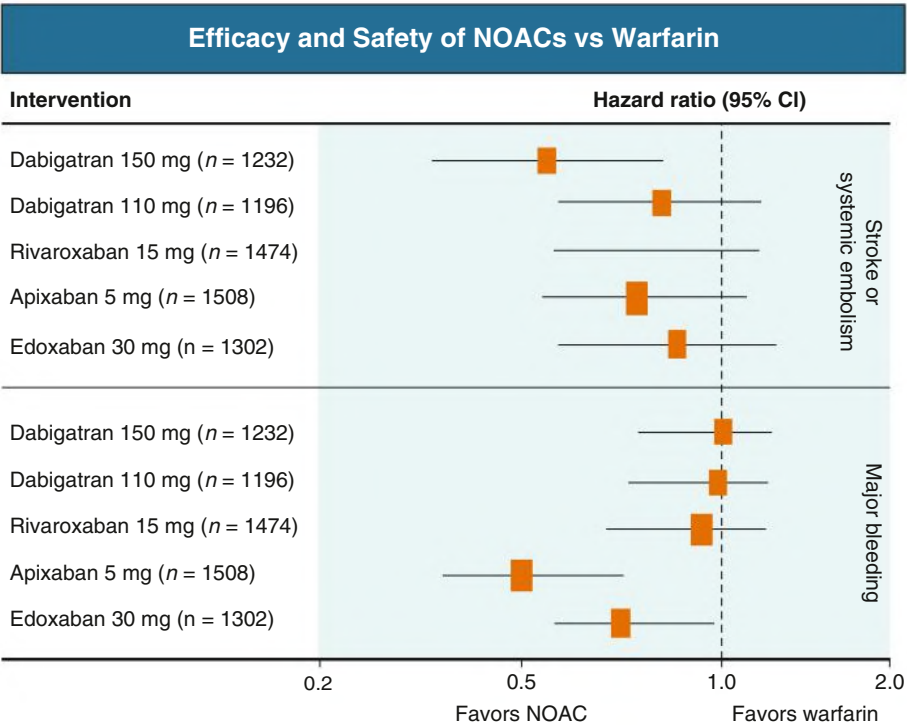


Fig. 85.6 Efficacy and safety of novel oral anticoagulants (NOACs) versus warfarin in the subgroup of patients with moderate chronic kidney disease from randomized, clinical trials in atrial fibrillation. CI, confidence interval. (From Qamar A, Bhatt DL. Stroke prevention in atrial fibrillation in patients with chronic kidney disease. *Circulation* 2016;133[15]:1512–1515.)

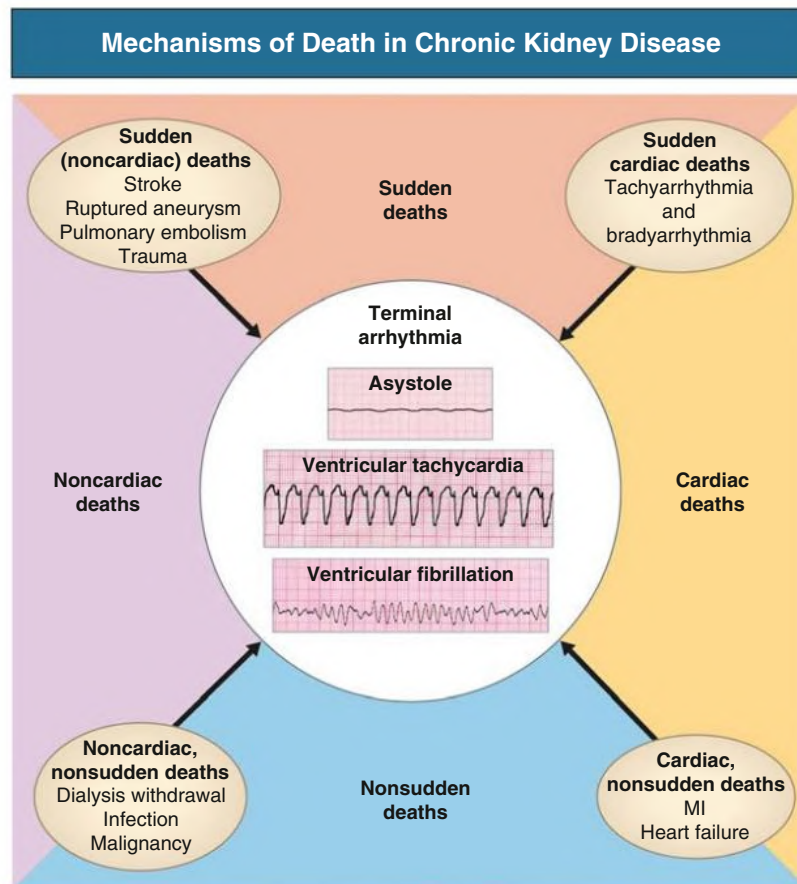


Fig. 85.7 Mechanisms of death in chronic kidney disease patients. *MI*, Myocardial infarction. (From Turakhia MP, Blankestijn PJ, Carrero JJ, et al. Chronic kidney disease and arrhythmias: conclusions from a Kidney Disease: Improving Global Outcomes (KDIGO) Controversies Conference. *Eur Heart J*. 2018;39:2314–2325.)

shifts and hyperkalemia in HD, autonomic dysfunction and sympathetic overactivity, including sleep apnea, and abnormalities in myocardial ultrastructure and function, including endothelial dysfunction, interstitial fibrosis, decreased perfusion reserve, and diminished ischemia tolerance.⁶² Low-potassium dialysate (<2 mmol/L) doubles the risk for cardiac arrest.⁶³ The rate of cardiac arrest is 50% higher for HD than for PD patients at 3 months after dialysis initiation but is higher for PD patients at 3 years. The highest rate of sudden cardiac death occurs in the first 2 months after HD initiation.

On-site defibrillation capability in HD centers (preferably with automatic external defibrillators) was recommended in a US practice guideline in 2005. The role that implantable cardioverter-defibrillators (ICDs) may play in reducing mortality in CKD patients is controversial, particularly regarding primary prevention. CKD may attenuate the survival advantage of ICDs, but older age and medical comorbidity should not routinely exclude patients from receiving ICDs. Despite the small number of dialysis patients in clinical studies, guidelines also recommend primary prophylactic ICD implantation if the ejection fraction is 35% or less. A small randomized controlled trial for prevention of sudden cardiac death in dialysis patients with LVEF greater than 35%, the ICD-2 trial, was terminated for futility.⁶⁴

VALVULAR DISEASE

The progression of calcific aortic valve stenosis is approximately three times faster in dialysis patients than in the general population (Fig. 85.8).⁶⁵ Annual echocardiography is recommended for asymptomatic dialysis patients with an aortic valve orifice area of less than 1.0

Prevalence of Valvular Calcification in the General Population and CKD

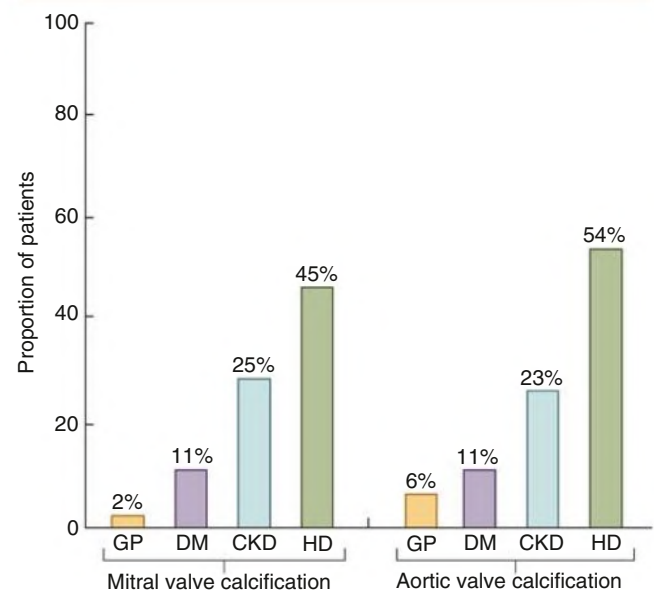


Fig. 85.8 Prevalence of valvular calcification in the general population and in patients with kidney disease. *CKD*, Chronic kidney disease; *DM*, diabetes mellitus; *GP*, general population; *HD*, hemodialysis. (From Marwick TH, Amann K, Bangalore S, et al. Chronic kidney disease and valvular heart disease: conclusions from a Kidney Disease: Improving Global Outcomes [KDIGO] Controversies Conference. *Kidney Int*. 2019;96:836–849.)

cm² who are suitable candidates for valve replacement. A meta-analysis of dialysis patients undergoing valve replacement surgery found no difference in survival for patients receiving tissue versus mechanical valves and fewer valve-related complications with tissue valves. The overall mortality is high, with in-hospital mortality about 20% (four times higher than in non-CKD patients) and a 2-year survival of 40%.⁶⁶ In kidney transplant recipients, in-hospital mortality was 11% for tissue and 15% for mechanical valve patients, and 2-year mortality rates were 62%

and 60%, respectively. In the entire cohort of kidney transplant patients, the rate of endocarditis after valve surgery was 5% per year.²³

Transcatheter aortic valve replacement may be appropriate in dialysis patients with symptomatic aortic stenosis who are not good candidates for surgery. Still, the prognosis of dialysis patients following transcatheter aortic valve replacement is worse compared with nondialysis patients.⁶⁷

SELF-ASSESSMENT QUESTIONS

1. SGLT2 inhibitors have not been shown to reduce the risk of:
 - A. hospitalization for heart failure.
 - B. cardiovascular death.
 - C. progression of kidney failure.
 - D. myocardial infarction.
 - E. progression to ESKD.
2. Strong predictors of sudden cardiac death in hemodialysis patients do not include:
 - A. a history of CHD.
 - B. a history of peripheral arterial disease.
 - C. diabetes.
 - D. reduced ejection fraction.
 - E. a history of atrial fibrillation.
3. CKD patients with an estimated glomerular filtration rate less than 30 mL/min/1.73 m² should be treated to achieve the guideline goal for secondary prevention for low-density lipoprotein cholesterol of:
 - A. less than 100 mg/dL.
 - B. less than 70 mg/dL.
 - C. less than 55 mg/dL.
 - D. less than 40 mg/dL.
 - E. less than 25 mg/dL.
4. Coronary artery disease in patients with CKD differs from non-CKD controls by exhibiting:
 - A. more and larger plaques.
 - B. fewer signs of inflammation.
 - C. less calcification.
 - D. fewer macrophages.
 - E. fewer T-lymphocytes.

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